Vision-Language-Action (VLA) Models for Industrial Robotic Applications

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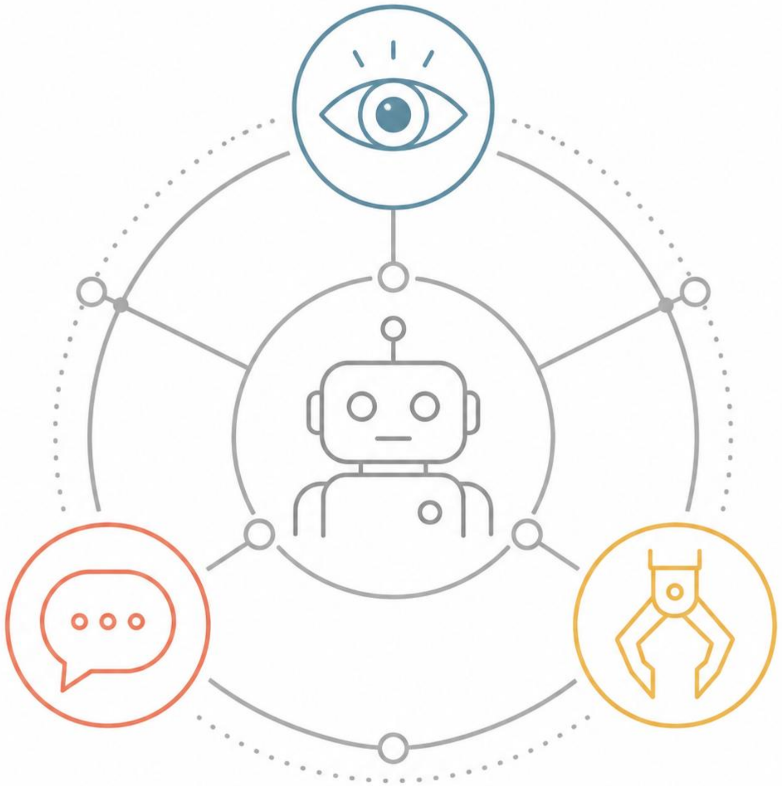


Agenda

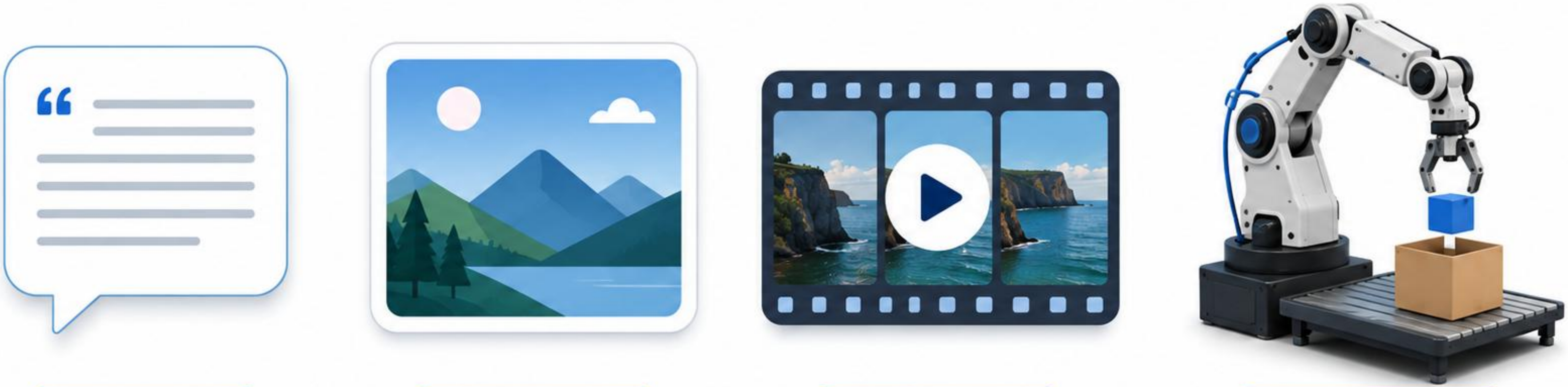
- From Generative AI to Generative Action
- Vision-Language-Action Models
- Capabilities and Limitations
- Challenges in Industrial Scenarios
- Our Research
- Outlook & Takeaways
- Q&A + Discussion

From Generative AI to Generative Action

A progression from understanding and creation to real-world execution

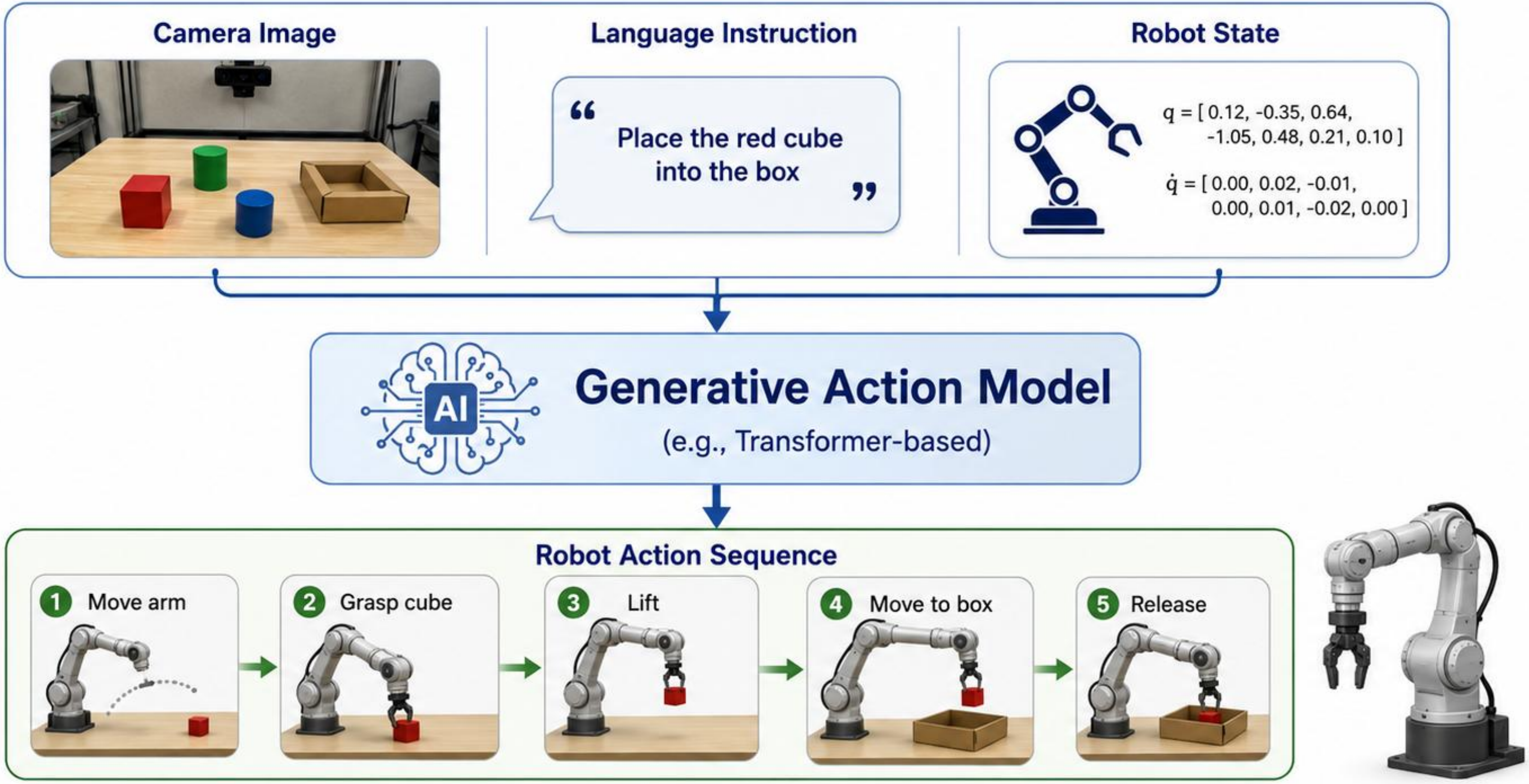
A horizontal progression:

1 Text → 2 Images → 3 Video → 4 Actions



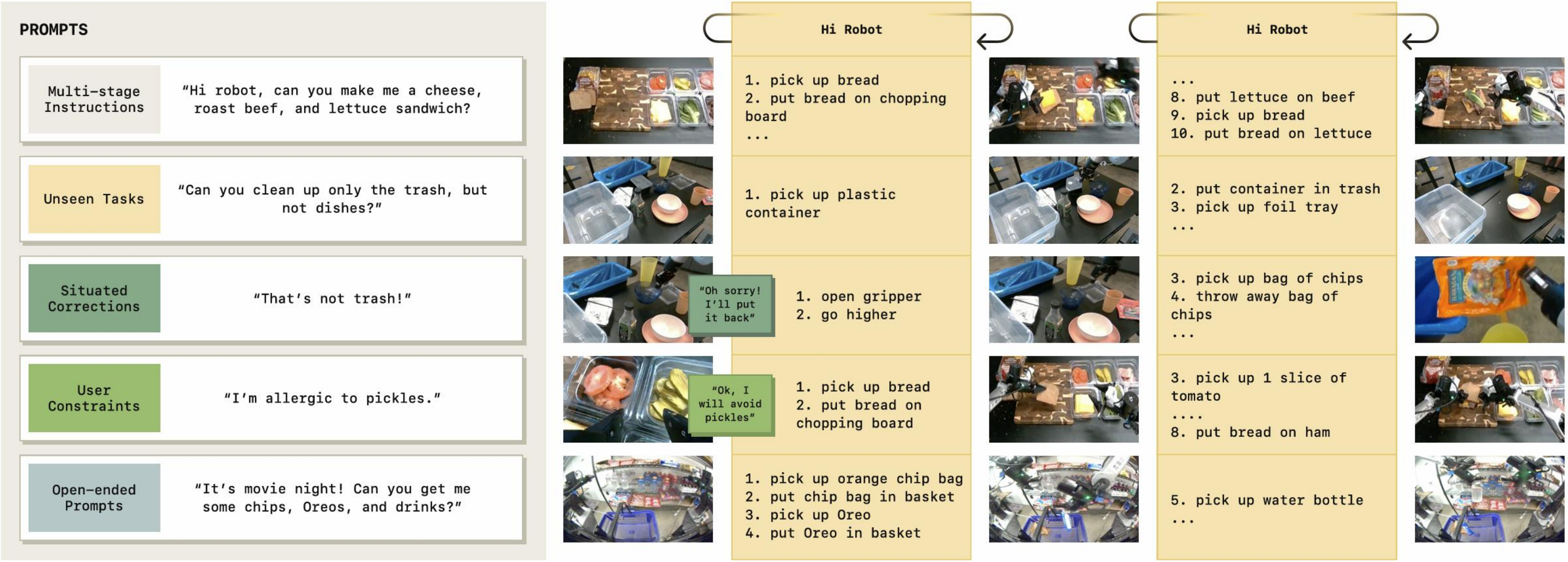
ChatGPT | DALL·E / Midjourney | Sora / Video Models | Robot Action Models

Generative Action Models: The Basic Idea



The model generates an action trajectory conditioned on visual observations, language goals, and robot state.

Example: Task execultion by a VLA

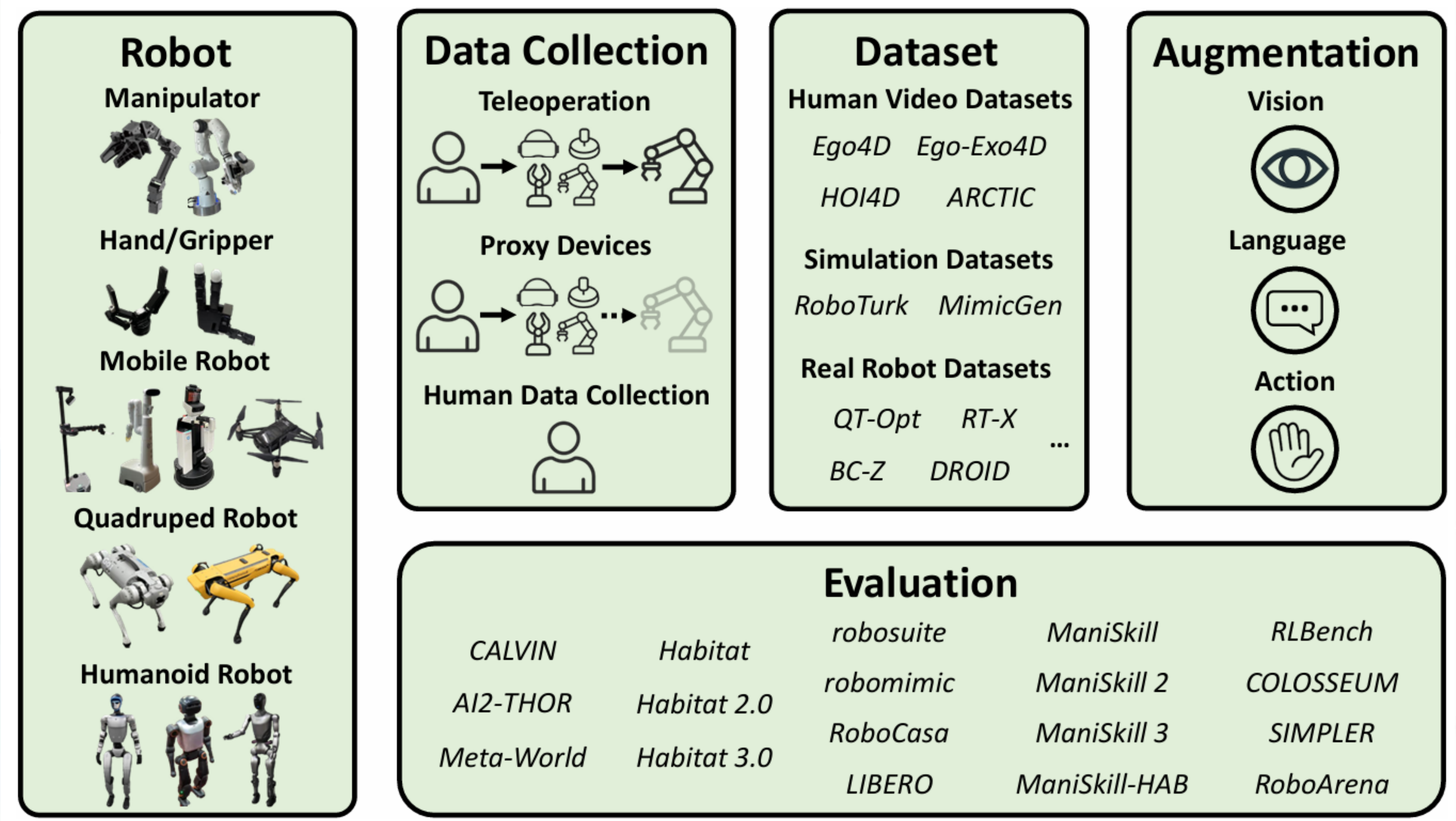


Video Examples:

https://www.pi.website/research/hirobot

Shi et al., Hi Robot: Open-Ended Instruction Following with Hierarchical Vision-Language-Action Models, Proceedings of ICML 2025, PMLR 267

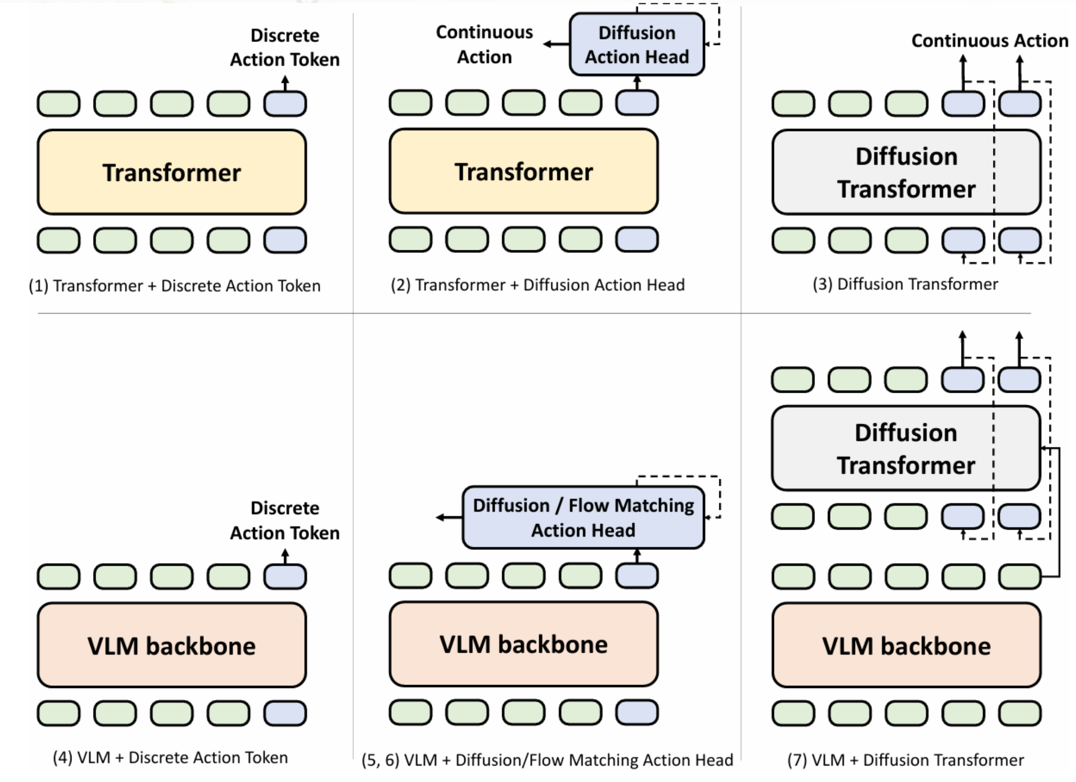
What is a Vision-Language-Action Model?



More data +
Larger models +
More diverse tasks
↓
Better representations
↓
Better generalization and robustness
↓
Toward generalist robot behavior

Kawaharazuka et. al, Vision-Language-Action Models for Robotics: A Review Towards Real-World Applications, IEEE Access 2025

From Imitation Learning to Vision-Language-Action Models

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VLA = imitation learning + vision-language
understanding + generative action prediction
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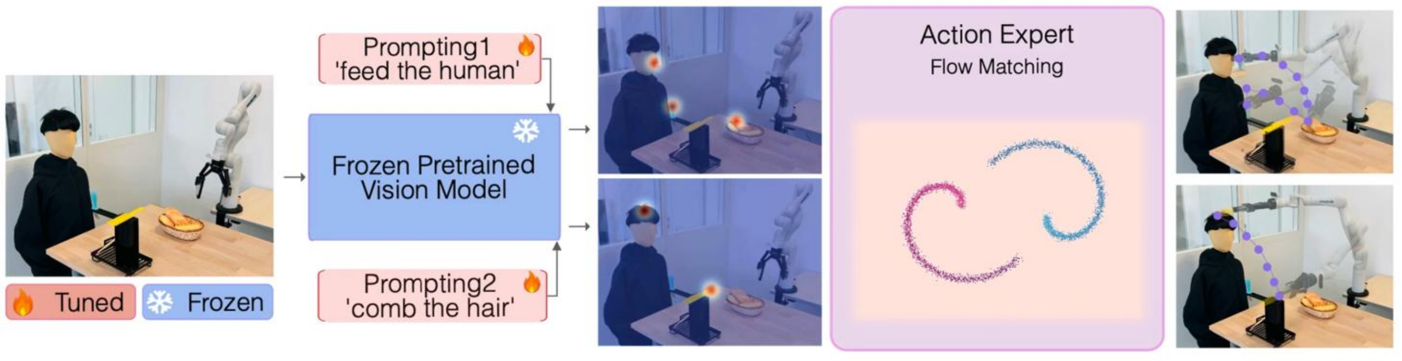
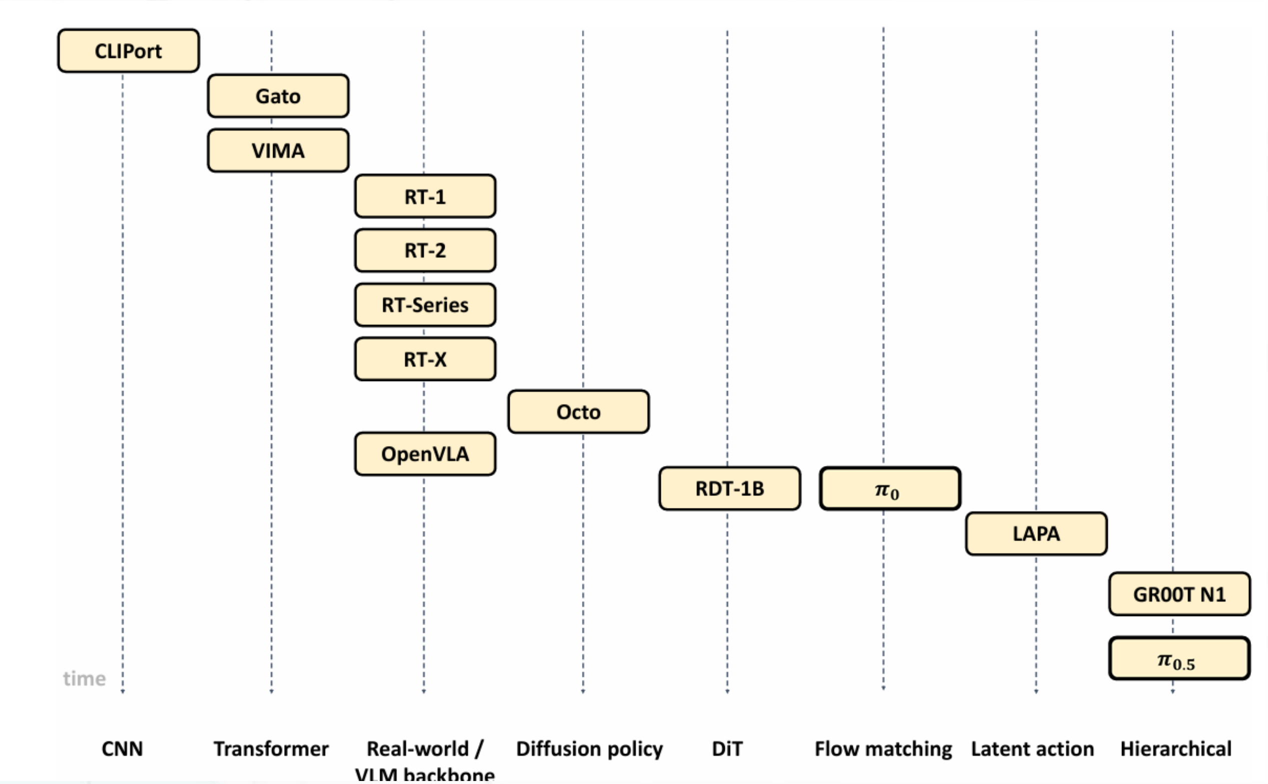


[Robot Manipulation with Flow Matching](),

Zhang and Gienger, *Affordance-based Robot Manipulation with Flow Matching, CoRRL 2024*

Kawaharazuka et al., Vision-Language-Action Models for Robotics: A Review Towards Real-World Applications, IEEE Access 2025

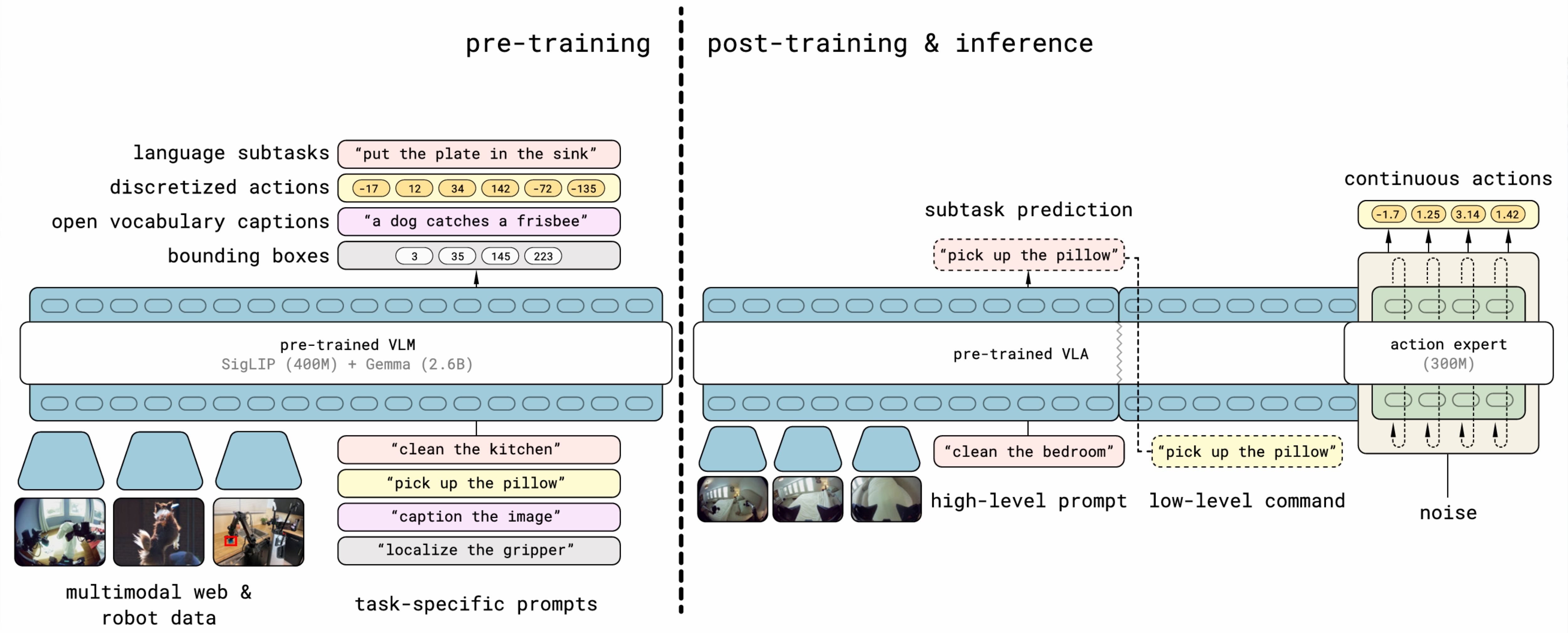
Timeline of VLA Development



Kawaharazuka et al., Vision-Language-Action Models for Robotics: A Review Towards Real-World Applications, IEEE Access 2025

Phase	Years	Main idea
Grounding and affordances	2021–2022	Connect language and vision to robot skills
Transformer robot policies	2022–2023	Scale imitation learning with large robot datasets
VLM-based VLAs	2023–2024	Use pretrained vision-language knowledge for robot action
Generative and deployable VLAs	2025–2026	Diffusion/flow actions, spatial grounding, real-time deployment, evaluation

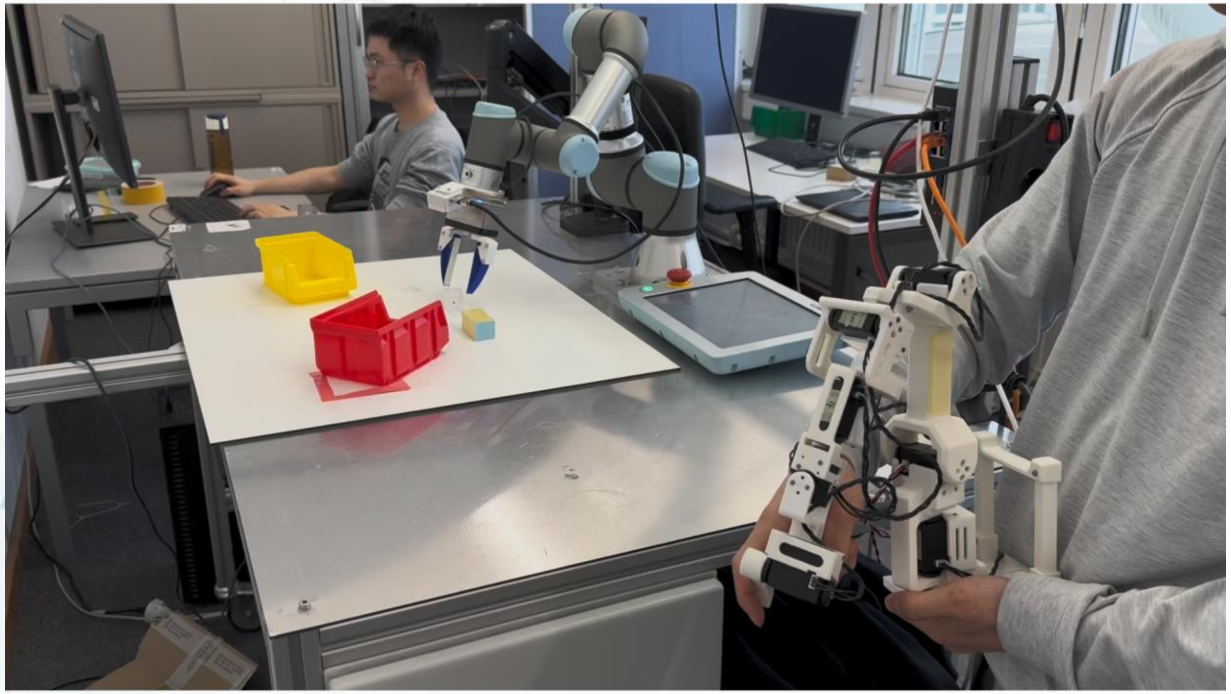
$\pi_{0.5}$: Model Architecture, Training and Inference



[Physical Intelligence, π_{0.5}: a Vision-Language-Action Model with Open-World Generalization]

Black et al., π0.5: a VLA with Open-World Generalization, CoRL 2025

Some Examples and Key Takeaways

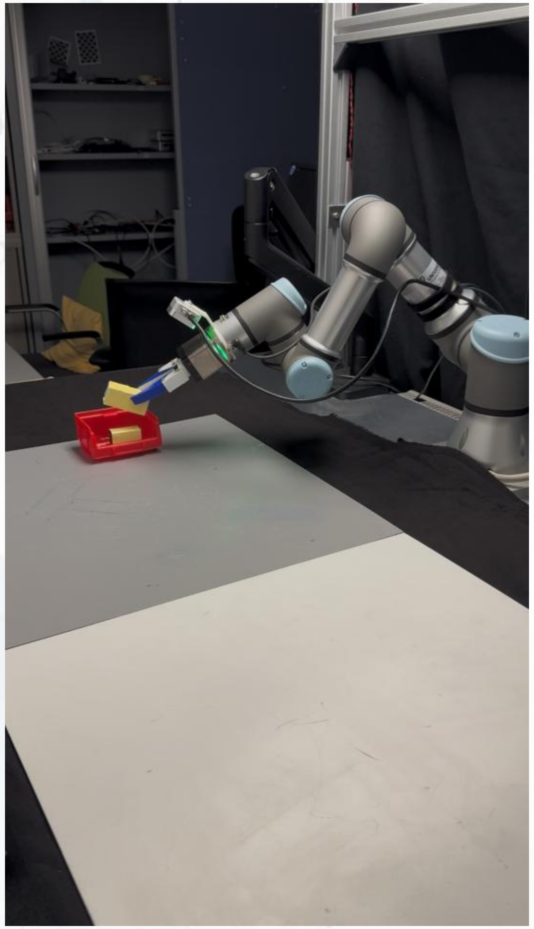


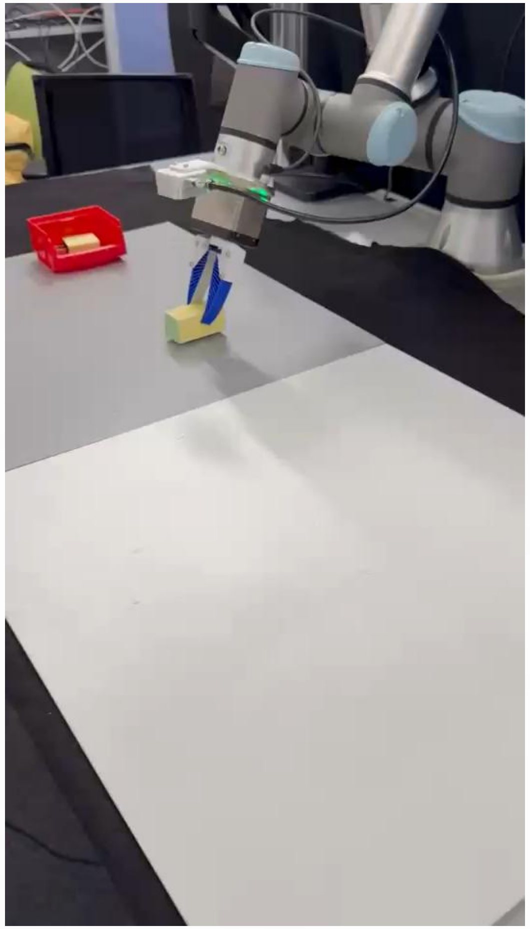
Does the model work out of the box?
- No, finetuning is necessary

What data do they need?
- Generally, teleoperation data to finetune the action head
- RGB data for the objects, visual surroundings etc.
- Language instruction connected with the actions

Can these be trained to perform multiple actions?
- Yes. More actions = More data = More compute

Some Examples and Key Takeaways





Does the Model perform well under changing conditions?

- Yes, generally, the model is robust to visual changes, object location changes and lighting conditions etc.

What are the ‚emergent abilities‛ or advantages over standard end-to-end learning approaches?

- Self-corrective behaviour
- Can perform in novel settings
- Multi-step actions

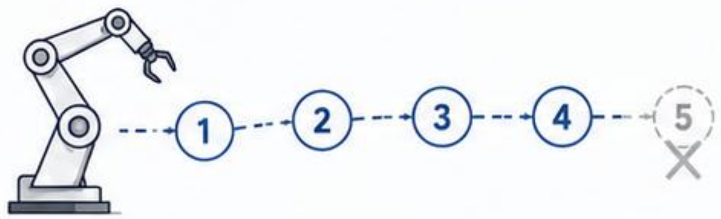
Some Examples and Key Takeaways

What are the limitations or downsides?
- End-to-end = Garbage in, Garbage out
- Autoregressive = Go from past to the future
- More challenging task = More data and compute needed
- Still poor at long-horizon tasks without guardrails
- Still poor at fine-tolerance tasks
- We are still restricted by the embodiment and the intelligence
- Not a panacea!

Current Limitations of Vision-Language-Action Models

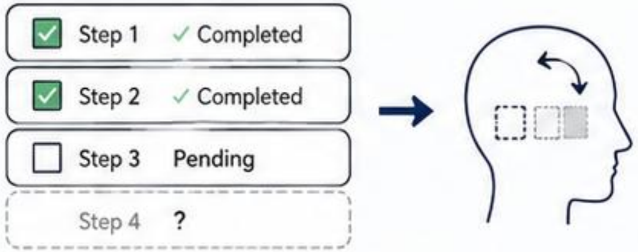
Why current VLA systems still struggle in real-world industrial workflows

1 Long-Horizon Execution



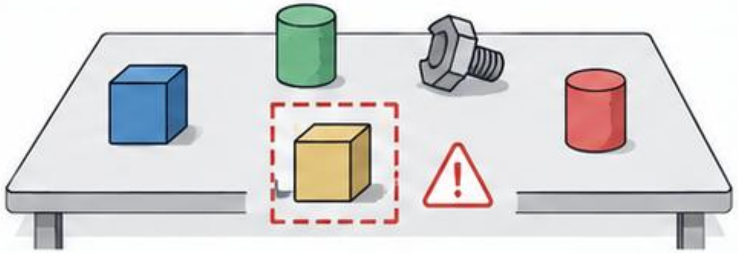
VLAs can perform short skills, but often lose task progress over long multi-step procedures.

2 Weak Task Memory



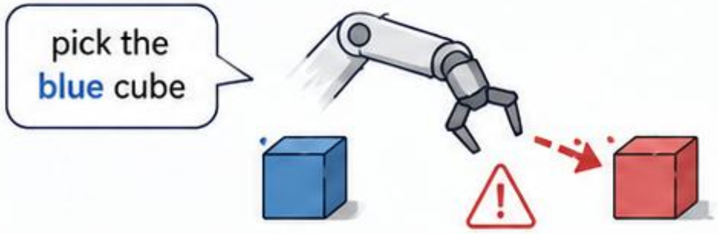
They may forget completed, pending, or failed steps without explicit task-state tracking.

3 Spatial Grounding Errors



Models may attend to the wrong object or region, especially in cluttered scenes.

4 Language Faithfulness



The robot does not always follow the instruction faithfully when multiple actions are possible.

5 Failure Detection & Recovery

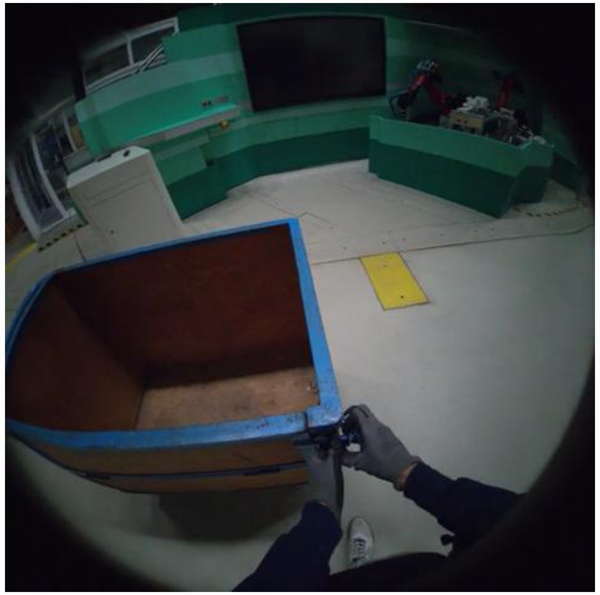
Current VLA systems often lack reliable postcondition checks, retry logic, and recovery behavior.

6 Robustness & Evaluation

Success on benchmarks does not guarantee robustness, safety, or reliability under realistic disturbances.

The field is moving beyond bigger pretrained models toward memory, grounding, controllability, robustness, and diagnostic evaluation.

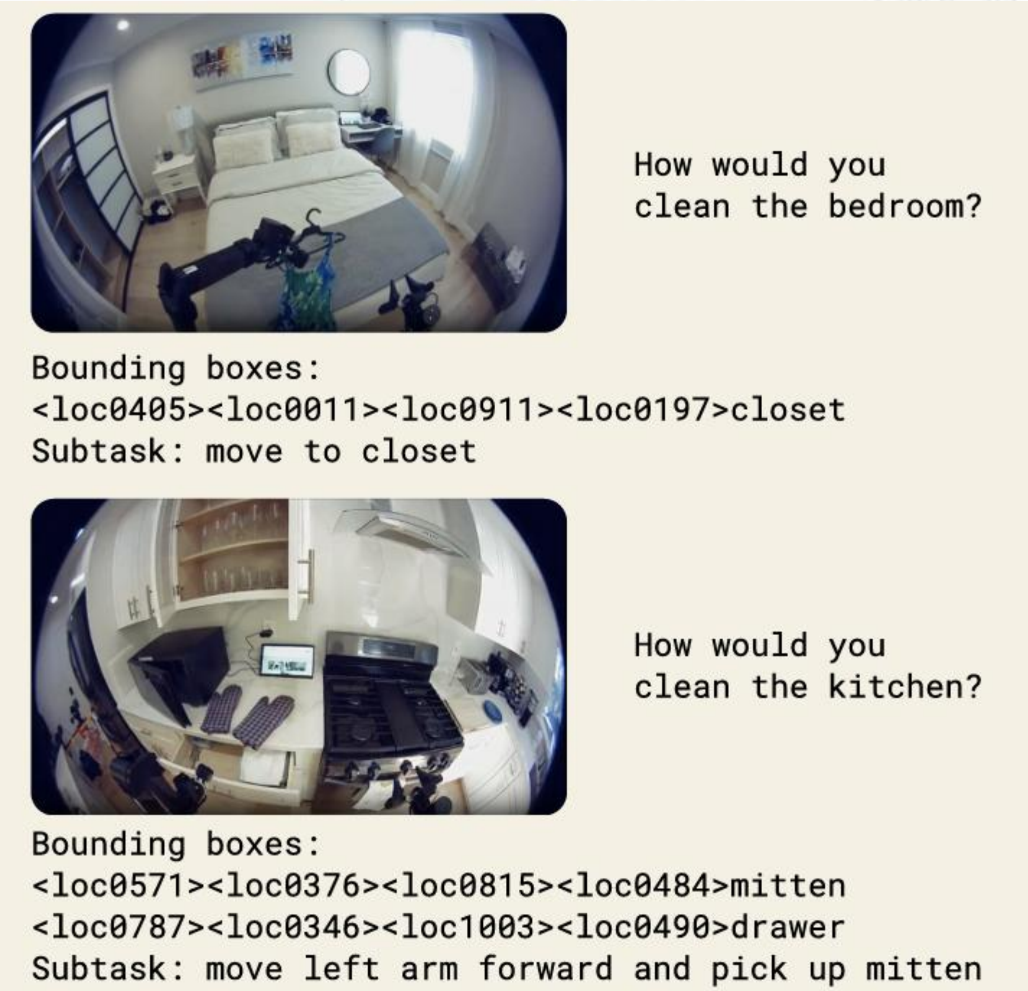
Egocentric-Exocentric Data & Benchmarks



- Egocentric data (w/ temporal annotations)
- Exocentric data
- Diverse industrial scenarios

IndEgo: A Dataset of Industrial Scenarios and Collaborative Work for Egocentric Assistants, NeurIPS 2025

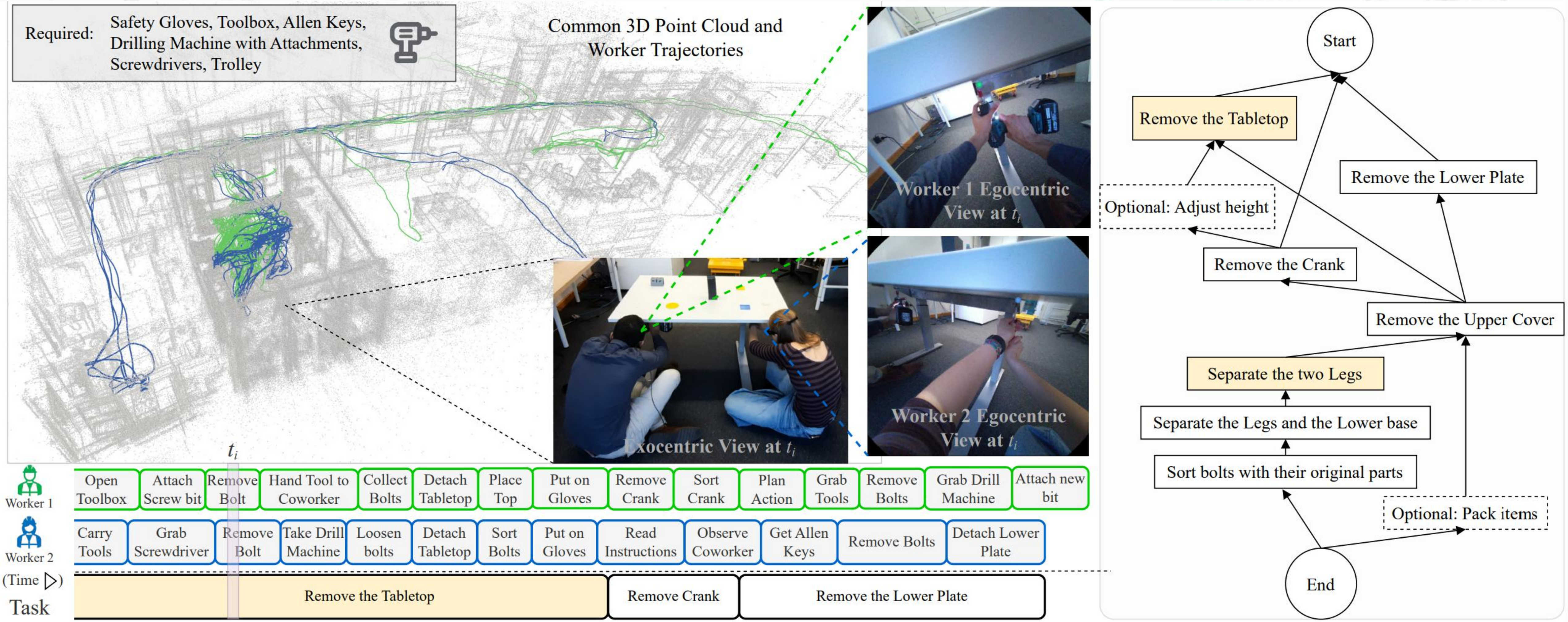
Black et al., π0.5: a VLA with Open-World Generalization, CoRL 2025



Egocentric/Exocentric data, video/action understanding is highly relevant to VLAs and real-world robotics

(pretraining, **policy adaptation, inference**)

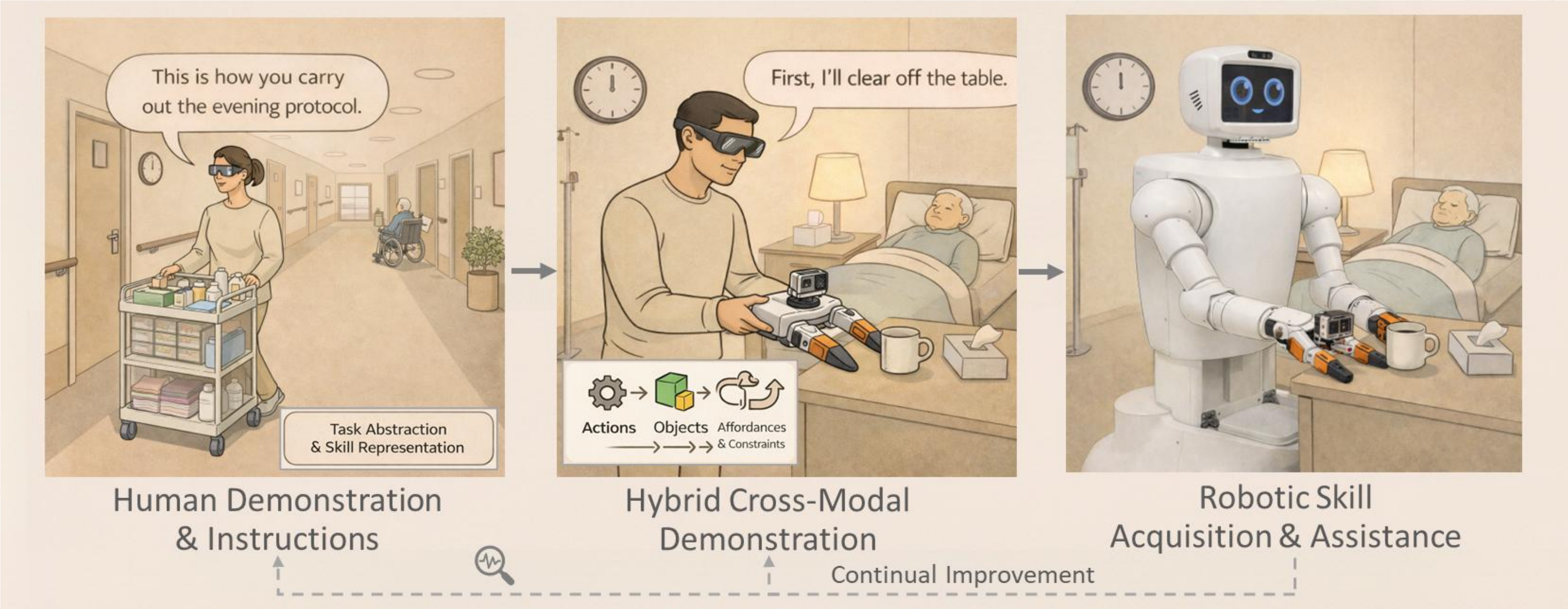
Structured Task Representations from Multimodal Data



IndEgo: A Dataset of Industrial Scenarios and Collaborative Work for Egocentric Assistants, NeurIPS 2025

Extracting user intent and context from Ego/Exo Demonstration

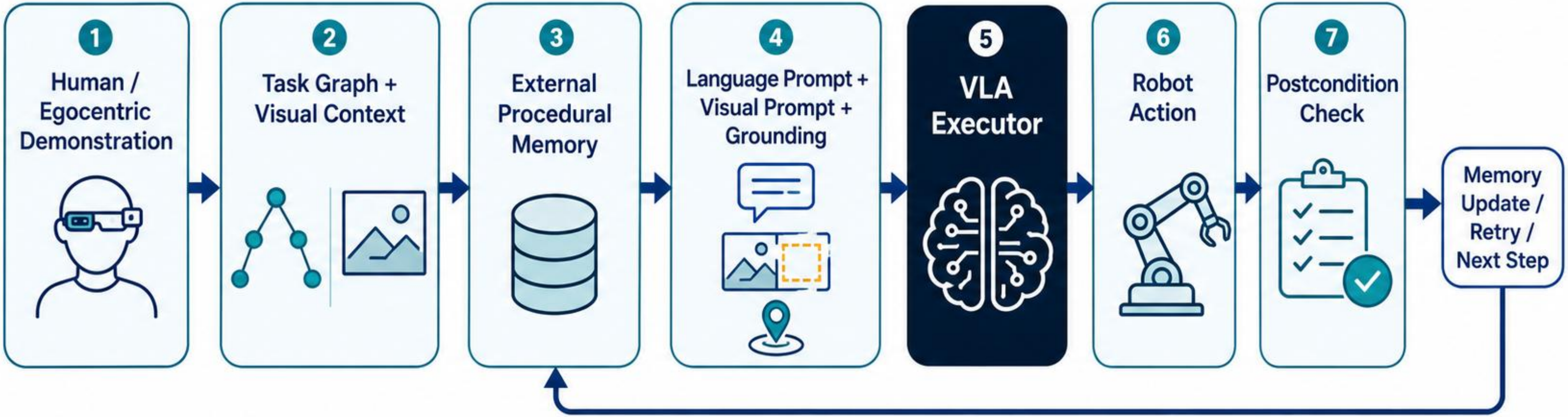
Hybrid Demonstration via a Universal Manipulation Interface



The MIMIC Pipeline for Service Robotics.
(Left) Context-Aware Instruction.
(Centre) Hybrid Demonstration.
(Right) Embodied Execution

Chi et al., Universal Manipulation Interface: In-The-Wild Robot Teaching Without In-The-Wild Robots, RSS 2024

MIMIC-VLA: Structured Context Around a VLA



- **Task graph:** defines valid steps and dependencies.
- **External memory:** tracks current, completed, pending, and failed steps.
- **Visual prompts:** highlight object, target region, or desired state.
- **Postcondition checks:** decide whether to continue, retry, or stop.

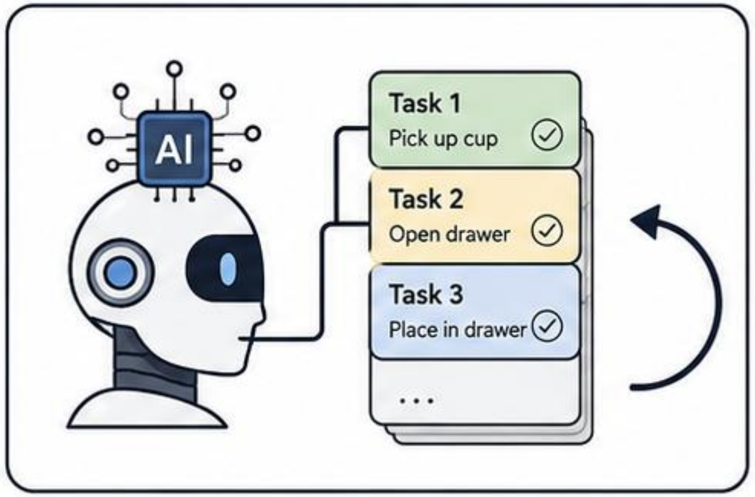
Why External Procedural Memory?

Internal memory helps the model remember. External memory helps the system control execution.

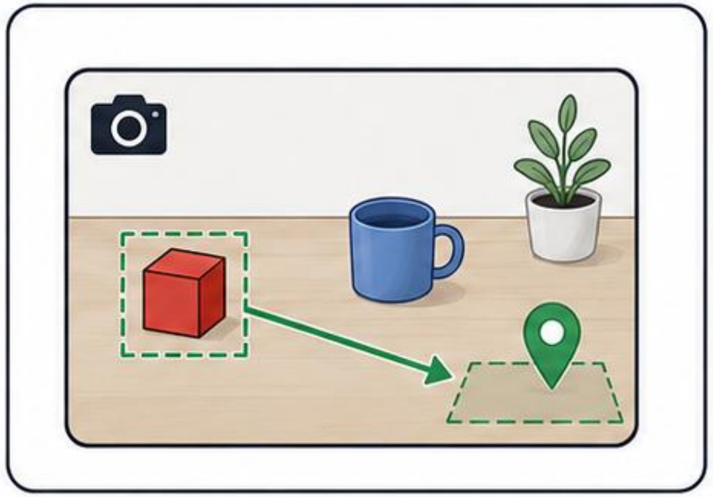
Internal VLA Memory	External Procedural Memory
Hidden inside the model	Explicit and inspectable
Learns temporal history	Tracks task state
Hard to edit	Human-editable
Helps perception/action continuity	Helps process control
Model decides what matters	Task graph defines what matters

External memory is not just remembering; it is procedural control.

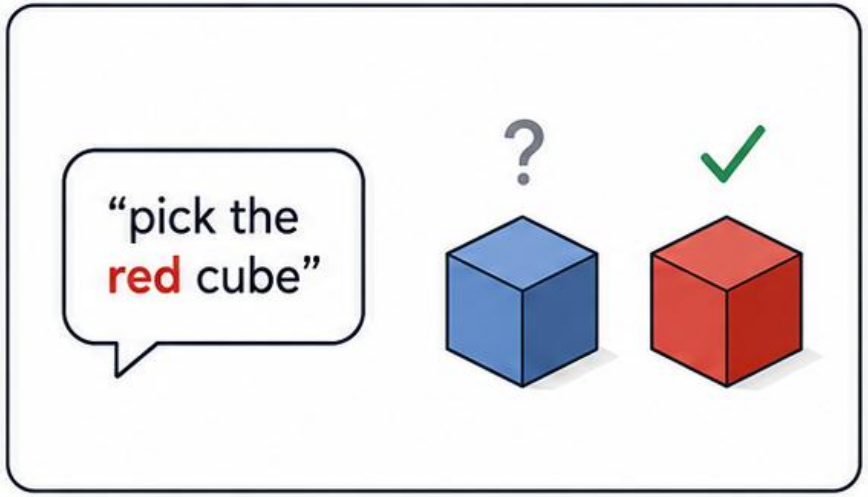
External memory acts as an execution-control layer: it selects the current step, chooses prompts, checks postconditions, and decides whether to continue, retry, or stop.



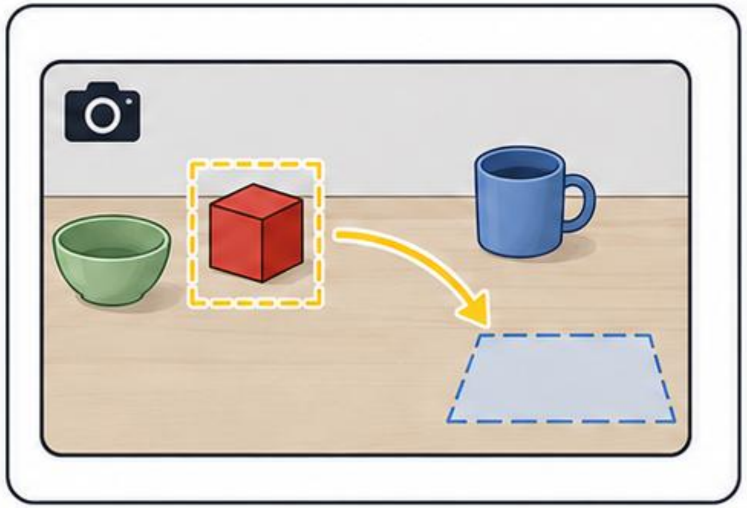
Memory



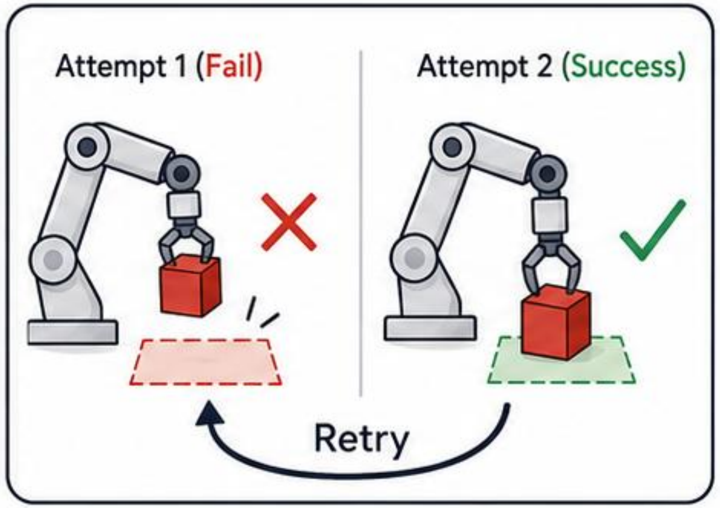
Spatial Grounding



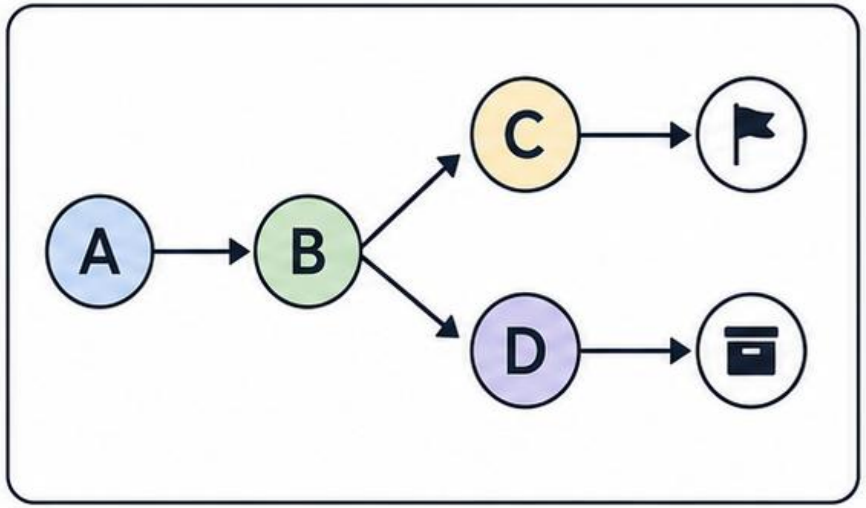
Language Faithfulness



Visual Prompting



Failure Detection



Task Graphs

Key Takeaways

1 VLAs connect perception, language, and action

Generate robot actions from images, instructions, and robot state.

2 Robot foundation models enable broader generalization

Larger models and broader data support more general robot behavior.

3 Manufacturing needs more than isolated skills

Industrial tasks require memory, task structure, grounding, and recovery.

4 Our research adds structure around the VLA

External procedural memory and visual prompts guide what the robot should do next.

5 Long-term vision: reliable industrial robot workflows

From isolated robot skills to reliable, structured industrial workflows.

The future is not only robots that can act, but robots that can follow structured procedures reliably.